

SAT Tactic Upgrade: The 1-Week Fix

Objective: Shift from “solving” to “mechanically dismantling” high-value trap questions. **Context:** You are losing points not because of a lack of knowledge, but because of specific, repeatable traps. These tactics are designed to be rigid procedures to bypass those traps.

Math Tactics

Tactic 1: The “No Real Solutions” Trigger

The Trap: You see a quadratic equation and the phrase “no real solutions,” and you try to solve for x or guess values. **The Fix:** STOP. Do not solve for x . 1. **Identify:** “No real solutions” = **Discriminant is Negative**. 2.

Formula: Write $b^2 - 4ac < 0$ immediately. 3. **Execute:** Plug in a, b, c and solve the inequality.

- **Example (Test 4, Q13):** $x^2 - 34x + c = 0$. No real solutions.
 - *Wrong Move:* Guessing c .
 - *Right Move:*
 - $a = 1, b = -34, c = c$.
 - $(-34)^2 - 4(1)(c) < 0$
 - $1156 - 4c < 0 \rightarrow 1156 < 4c \rightarrow 289 < c$.
 - Answer: 290 (or anything > 289).

Tactic 2: Vertex Form Translation

The Trap: You see a word problem about “maximum profit,” “minimum height,” or “reaches a peak,” and you try to model it with $y = mx + b$ or standard form. **The Fix:** “Max/Min” = **Vertex Form**. 1. **Template:** Write $y = a(x - h)^2 + k$. 2. **Fill Vertex:** (h, k) is the max/min point given in the text. 3. **Find a :** Plug in *any other point* (x, y) given in the text to solve for a . **Do not assume** $a = 1$ or $a = -1$.

- **Example (Test 4, Q22):** Vertex is $(9, -14)$. Passes through origin (implied or other point).
 - *Step 1:* $y = a(x - 9)^2 - 14$.
 - *Step 2:* Plug in other point to find a .
 - *Step 3:* Expand to standard form $ax^2 + bx + c$ only if requested.

Tactic 3: Dimensional Analysis for Area/Volume

The Trap: You convert units linearly ($1 \text{ ft} = 12 \text{ in}$) when dealing with Area (ft^2) or Volume (ft^3). **The Fix:** **Square the Factor**. 1. **Identify:** Is it Area (2) or Volume (3)? 2. **Modify Factor:** If 1 unit A = k unit B... * Area: 1 unit $A^2 = k^2$ unit B^2 . * Volume: 1 unit $A^3 = k^3$ unit B^3 .

- **Example (Test 9, Q21):** Convert $x \text{ m}^2$ to ft^2 where $1 \text{ m} \approx 3.28 \text{ ft}$.
 - *Wrong Move:* Multiply by 3.28.
 - *Right Move:* Multiply by $(3.28)^2 \approx 10.76$.
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Reading & Writing Tactics

Tactic 4: The "Slash & Burn" for Grammar

The Trap: You read the sentence "by ear" to see where the pause falls. **The Fix: Cross out the junk.** 1.

Prepositional Phrases: Cross out anything starting with *of, in, at, for, by, on, with*. 2. **Appositives:** Cross out the fluff between commas (e.g., ", a famous painter,"). 3. **Check Match:** Match the remaining Subject to the Verb.

- **Example (Test 8, Q20):** "The shape of the wings... [is/are]..."
 - *Slash:* "The shape ~~of the wings~~..."
 - *Result:* "The shape... **is**." (Singular).

Tactic 5: Scientific Logic Mapping ($A \rightarrow B$)

The Trap: You get overwhelmed by jargon ("piezophiles," "TMAO," "Neronian tools") and pick an answer that repeats keywords but gets the logic wrong. **The Fix: Simplify to Variables.** 1. **Ignore Nouns:** Call them X and Y . 2. **Map the Arrow:** Does X cause Y ? Does X prevent Y ? 3. **Identify the Gap:** The question usually asks for support. Find the choice that strengthens the arrow.

- **Example (Test 7, Q12 - TMAO):**
 - *Text:* Pressure compresses water ($P \rightarrow C$). TMAO is high in deep fish. Hypothesis: TMAO prevents compression ($T \rightarrow \text{No } C$).
 - *Task:* Support the hypothesis.
 - *Target:* We need evidence that **TMAO stops compression**.
 - *Choice D:* "Hydrogen bonds are more stable (less compressed) when TMAO is present." $\rightarrow$ Matches $T \rightarrow \text{No } C$.

Tactic 6: The "Prompt First" Protocol (Rhetorical Synthesis)

The Trap: You read the bullet points, understand the facts, and pick the answer that summarizes them best.

The Fix: Read the Question Sentence FIRST. 1. **Ignore Bullets:** Jump straight to "The student wants to..." 2.

Identify Constraint: Underline the specific goal (e.g., "emphasize a contrast," "explain the origin," "introduce the study"). 3. **Eliminate:** Cross out any answer that is true but ignores the constraint.

- **Example:** "The student wants to emphasize the **contrast** between the two theories."
 - *Choice A:* Theory X says this, and Theory Y says that. (No transition).
 - *Choice B:* **While** Theory X says this, Theory Y argues that. (Has "While" = Contrast).
 - *Select:* B.

Tactic 7: The "Full Stop" Check (Boundaries)

The Trap: You see a transition word like "however" or "therefore" and assume it needs a semicolon or comma based on "flow." **The Fix: The Independent Clause Test.** 1. **Test:** Can the part *before* the

punctuation stand alone as a sentence? Can the part *after* stand alone? 2. **Rule:** * Yes + Yes = **Period** (.) or

Semicolon (;). * Yes + No = **Comma** (,). * **CRITICAL:** "Semicolon + However + Comma" is a standard pattern for joining two full sentences. "; However,"

- **Example (Test 7, Q17):** "...capture its prey; rather, the brightly colored arachnid..."
 - *Check:* "capture its prey" (Sentence). "the brightly colored arachnid..." (Sentence).
 - *Fix:* Needs a hard stop. Semicolon is the only choice that acts like a period.